Extracting Commonsense Knowledge for Robotic Agents from LLMs

Mark Adamik, Ilaria Tiddi, Stefan Schlobach

K-CAP
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KCAI Knowledge in Artificial Intelligence

Motivation

Commonsense Reasoning for Robots

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Commonsense Reasoning for Robots

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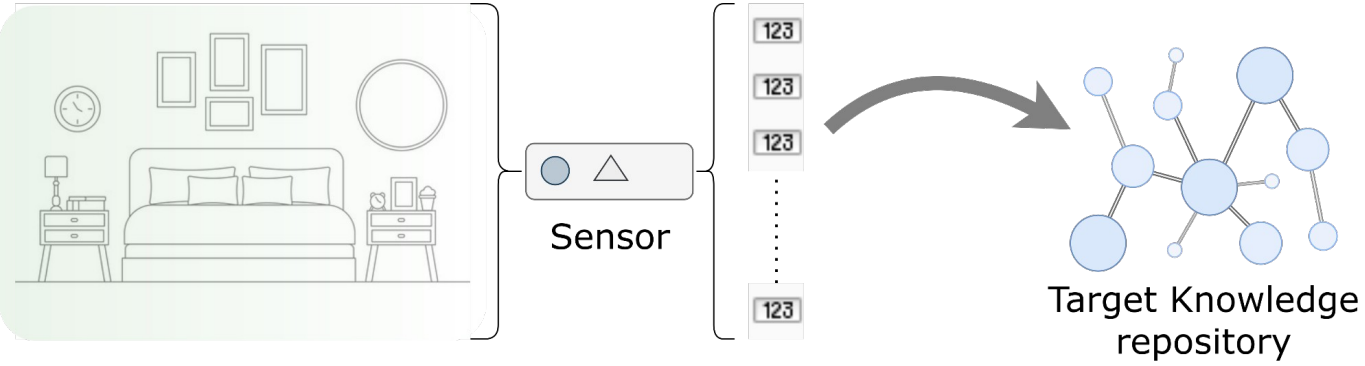
- VLAs are promising, but ultimately lacking safety guarantees
- For real world deployment safety and explainability are essential

↓

- Can the use of knowledge graphs help address this issue?

Overarching Research Problem

Given a **knowledge graph**, how can a robotic agent associate **sensory data** with **concepts** in these **knowledge graphs**?



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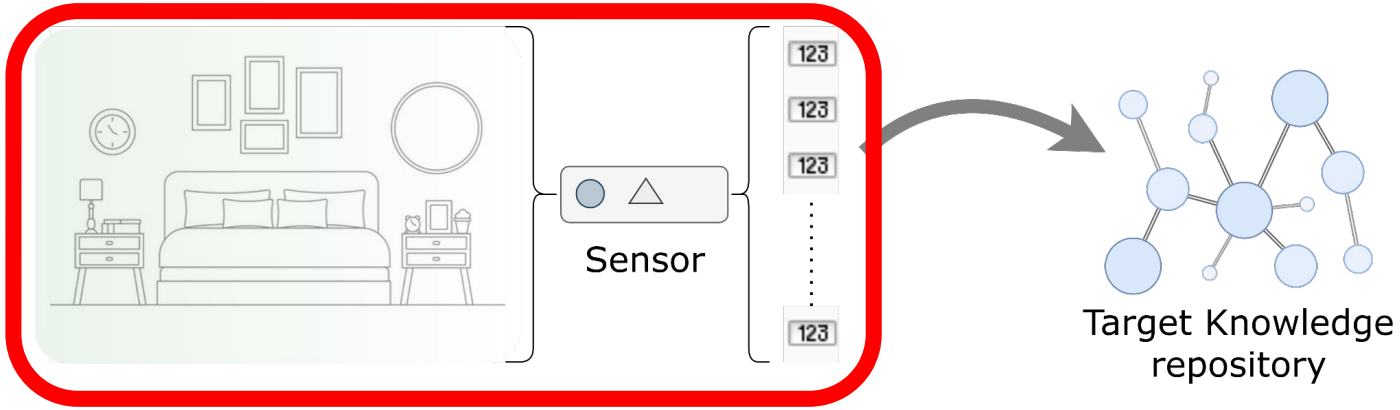

Adamik, M., Pernisch, R., Tiddi, I., & Schlobach, S. (2024, November). **ORKA: An ontology for robotic knowledge acquisition**. In International Conference on Knowledge Engineering and Knowledge Management

Adamik, M., Tiddi, I., Pernisch, R., & Schlobach, S. (2024, November). **Large-scale knowledge graphs as a tool for enhanced robotic perception**. In Proceedings of the AAAI Symposium Series

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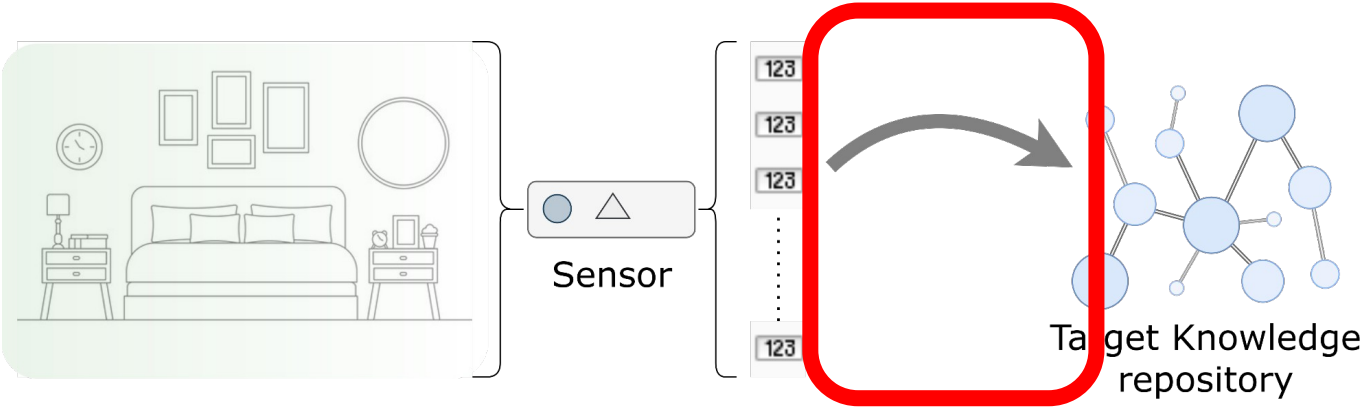

Adamik, M., Pernisch, R., Tiddi, I., & Schlobach, S. (2024, November). **Advancing robotic perception with perceived-entity linking** In International Semantic Web Conference
Adamik, M., & Schlobach, S. (2023, August). **Towards a Definition and Conceptualisation of the Perceived-Entity Linking Problem**. In *RobOntics@ RO-MAN*

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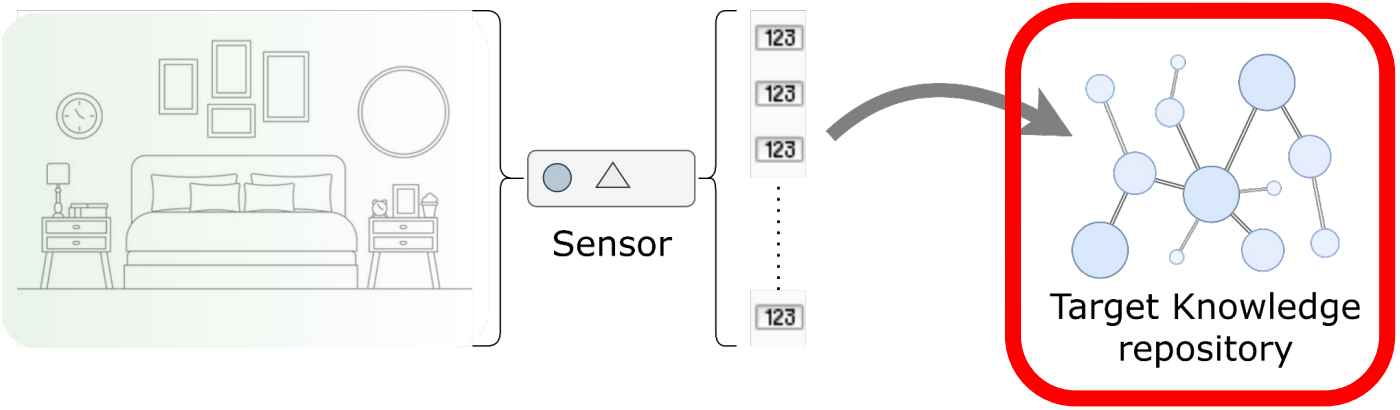

Sources of Commonsense Knowledge

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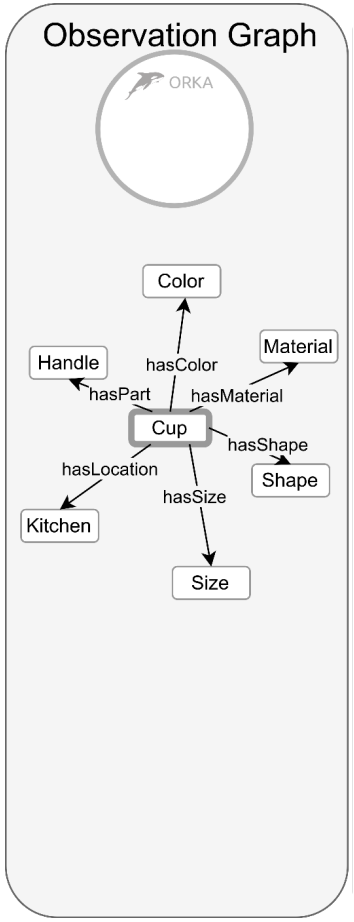
DBpedia

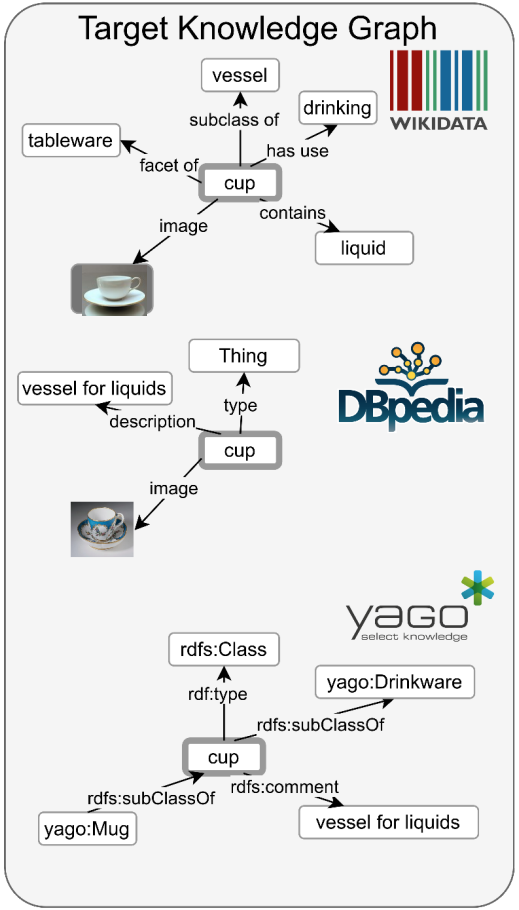
yago
select knowledge

WIKIDATA

RoboEarth

Example problem





Sources of Commonsense Knowledge

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“LLMs are trained on human commonsense knowledge.”

- Pascal Hitzler, 28 hours ago

Research Question

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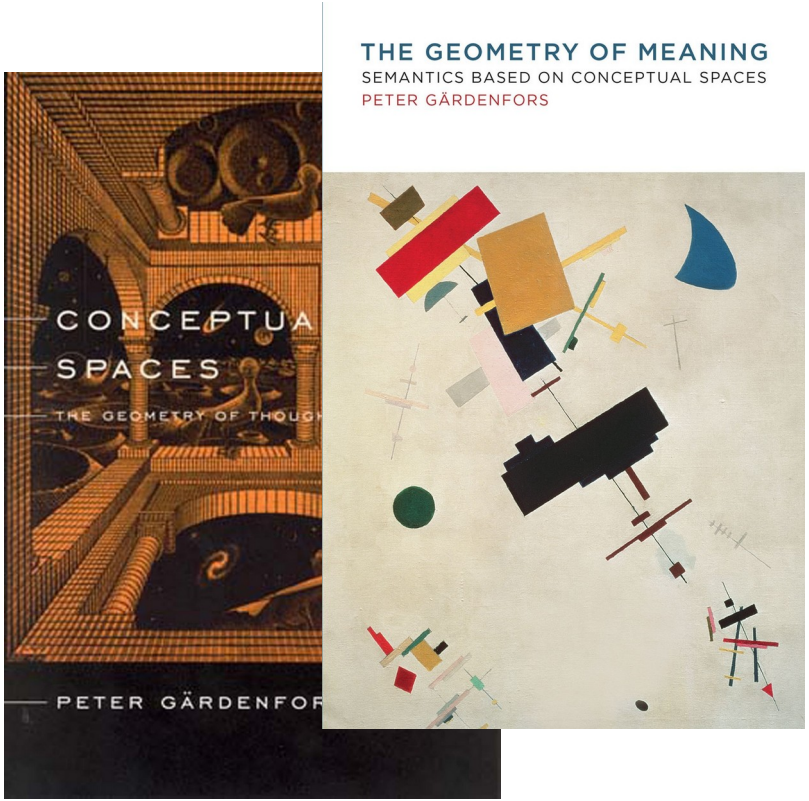
> *How can the **latent common-sense information** within **LLMs** can be **systematically extracted** and transformed into a **structured, symbolic format** suitable for use in reasoning and planning tasks?*

Approach

Background: Conceptual Spaces

What are conceptual spaces?

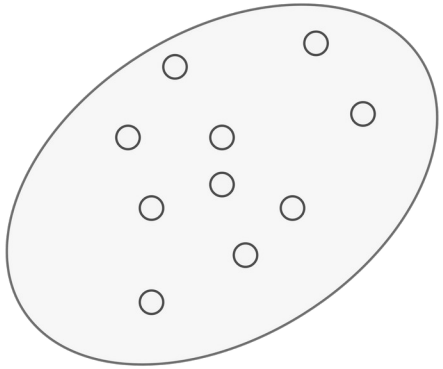
- Geometric representation format



Background: Conceptual Spaces

What are conceptual spaces?

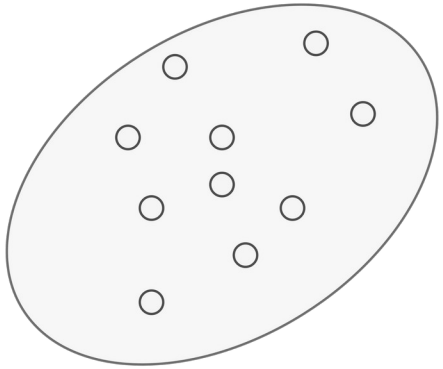
- Geometric representation format
- Convex shapes



Background: Conceptual Spaces

What are conceptual spaces?

- Geometric representation format
- Convex shapes
- Based on a number of **quality dimensions**



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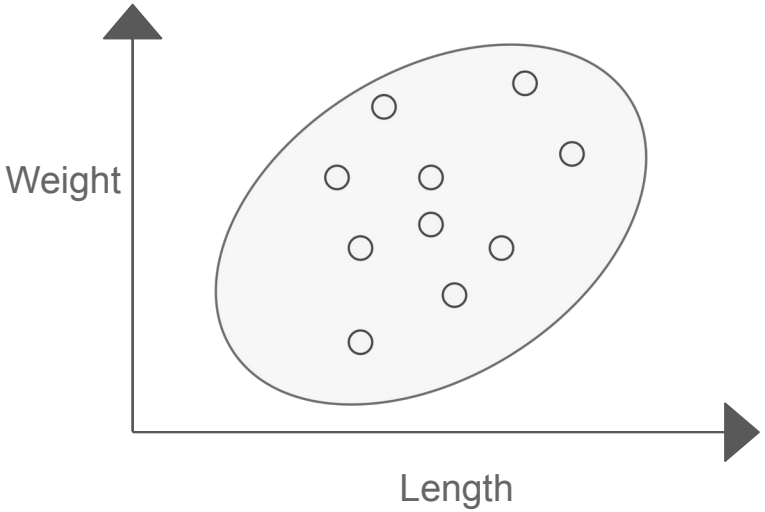

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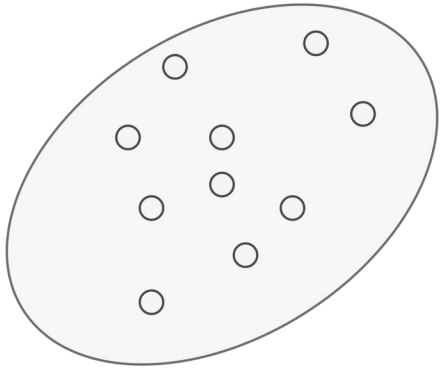

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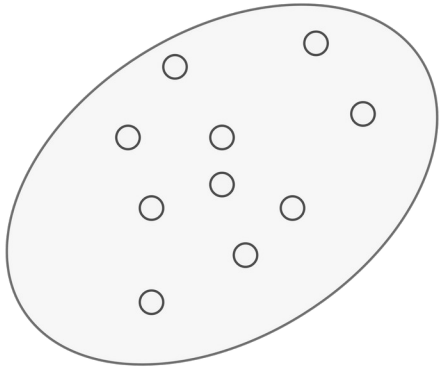

What are the benefits of using conceptual spaces?

- Intermediate layer between symbolic and connectionist representations
- Similarity could be represented (prototype theory [Rosch 1971])

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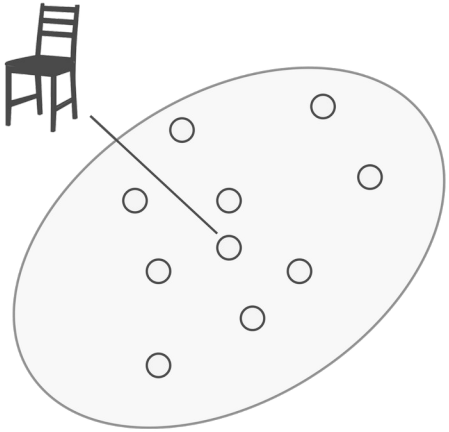

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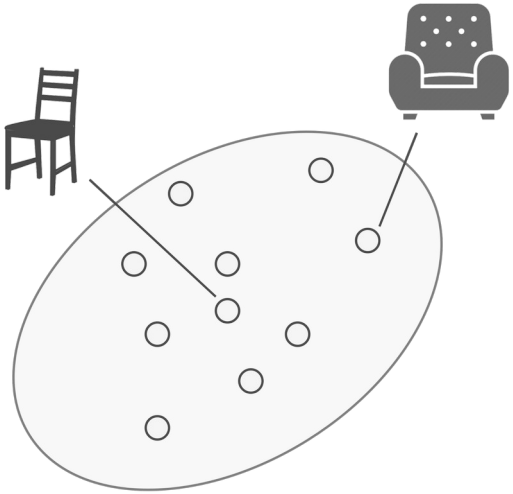

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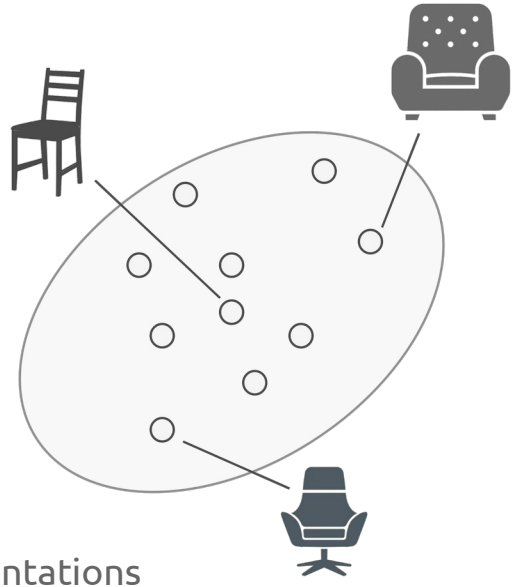

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What are the benefits of using conceptual spaces?

- Intermediate layer between symbolic and connectionist representations
- Similarity could be represented (prototype theory [Rosch 1971])
- (Hopefully) Explainable dimensions
- Objects (and changes to them) could be represented as a trajectory

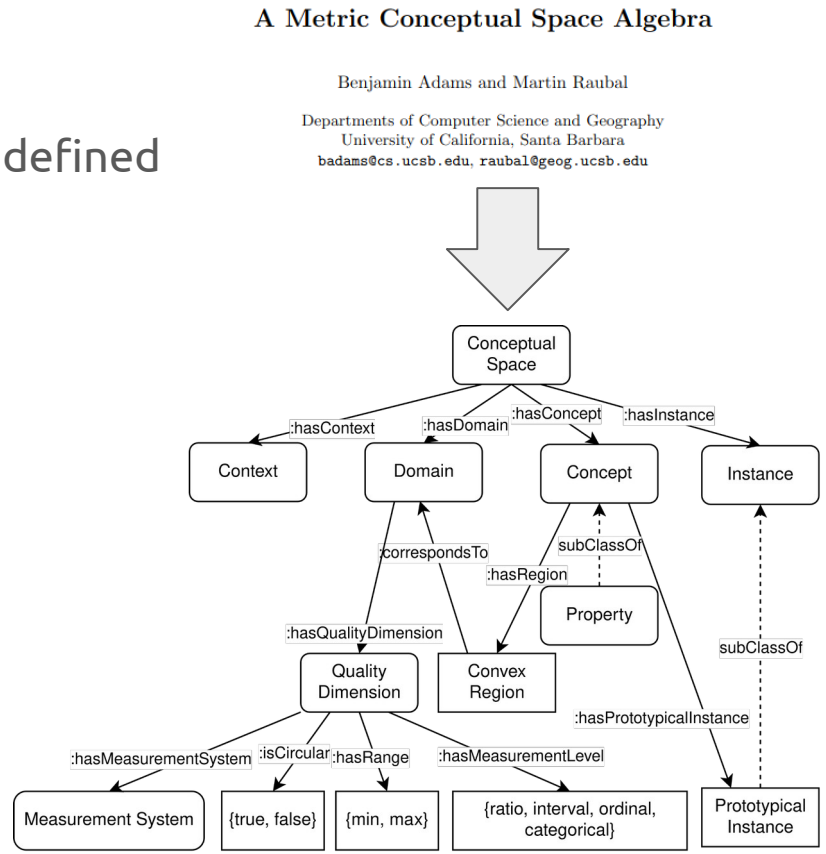
Generating a commonsense knowledge graph - Idea

Inspired by Conceptual Spaces

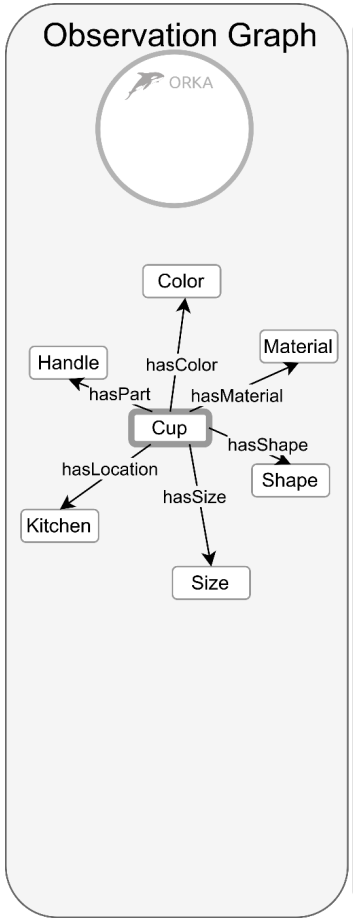
Task: extract prototypical instances of objects defined along a set of domains:

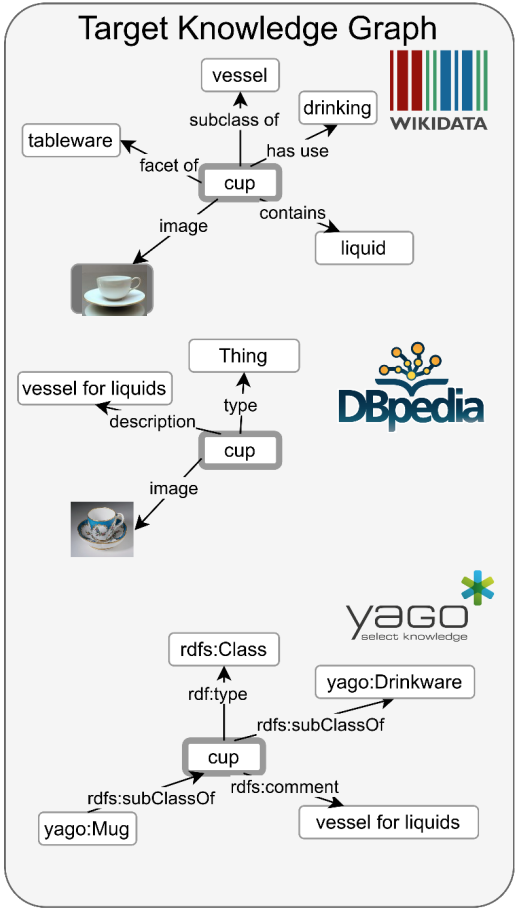
- Numerical
  - Size (Length, Width, Height)
  - Temperature
  - Weight

- Categorical
  - Shape
  - Material
  - Location
  - Color
  - Disposition



Example problem





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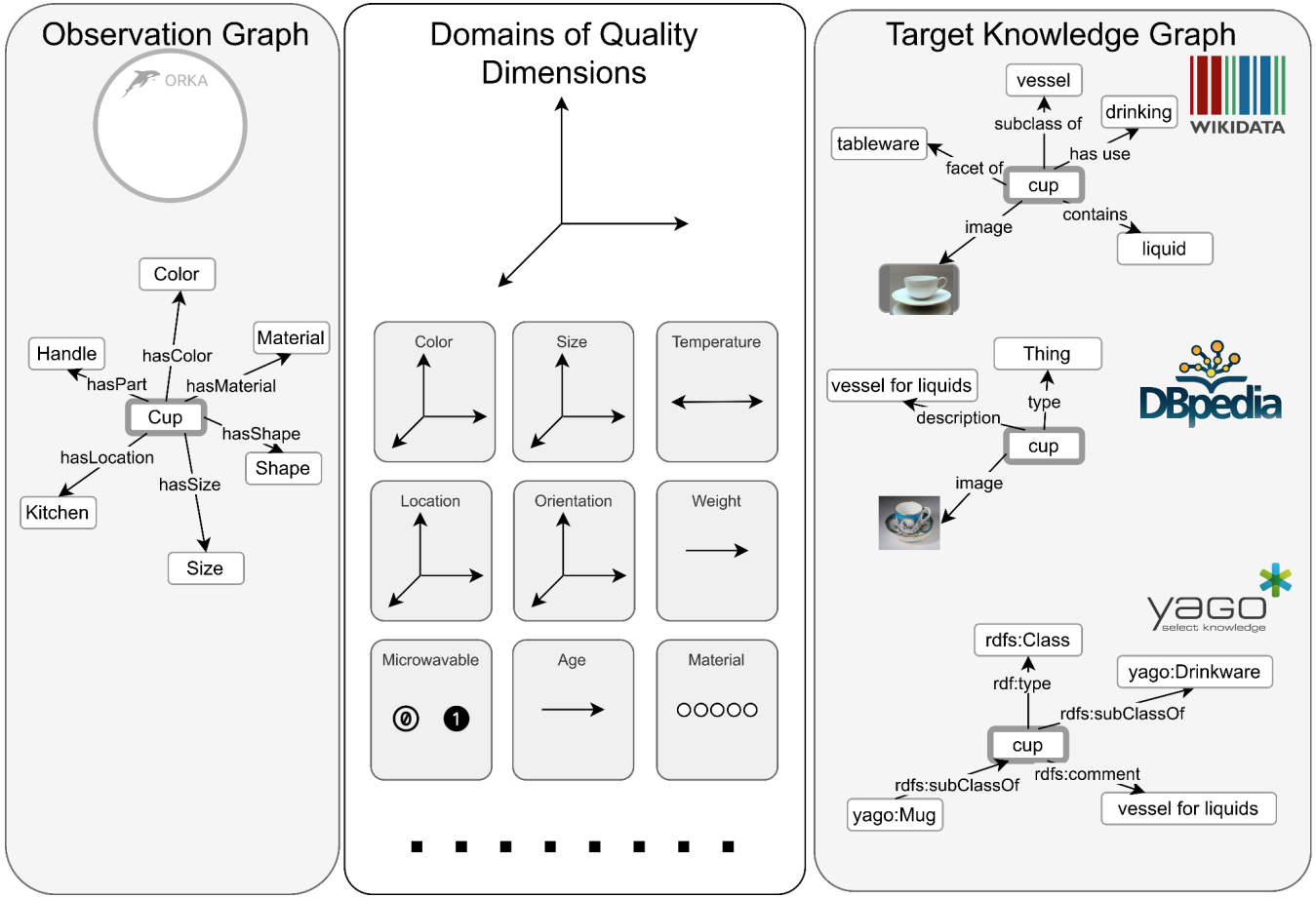
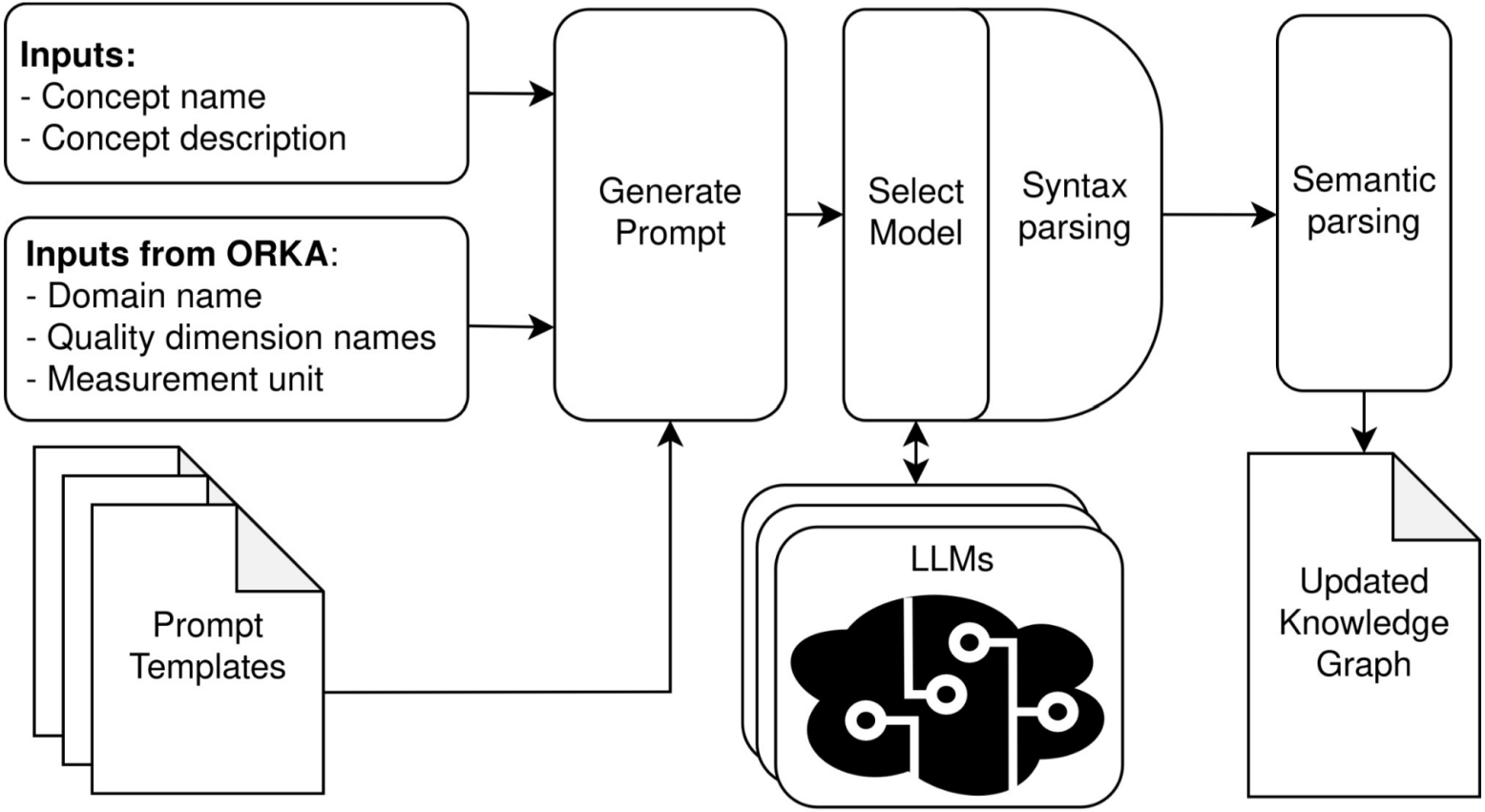
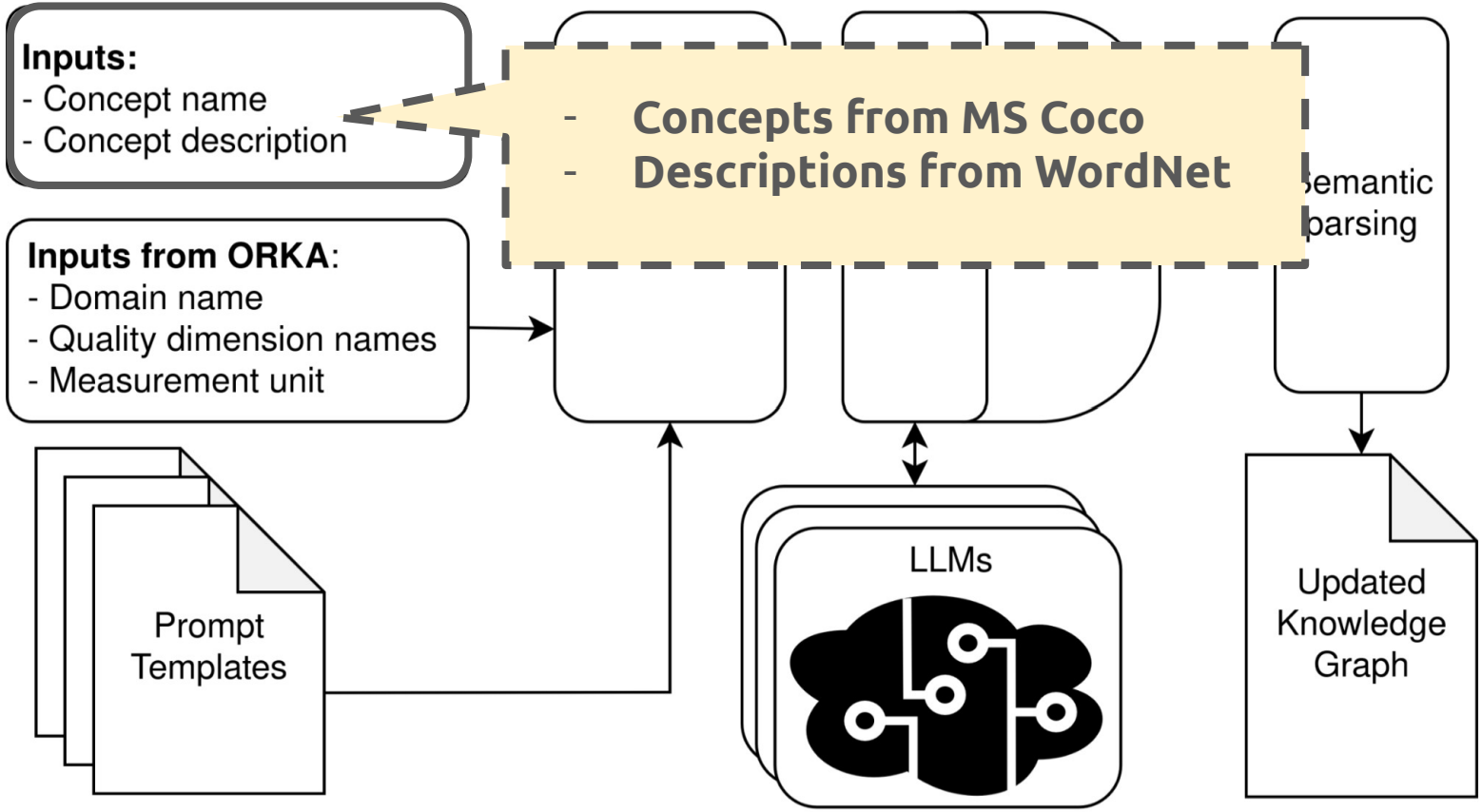


Pipeline


Inputs:
- Concept name
- Concept description
Inputs from ORKA:
- Domain name
- Quality dimension names
- Measurement unit
Generate Prompt
Select Model
Syntax parsing
Semantic parsing
Prompt Templates
LLMs
Updated Knowledge Graph

Pipeline


Inputs:
- Concept name
- Concept description
- Concepts from MS Coco
- Descriptions from WordNet
Inputs from ORKA:
- Domain name
- Quality dimension names
- Measurement unit
Prompt Templates
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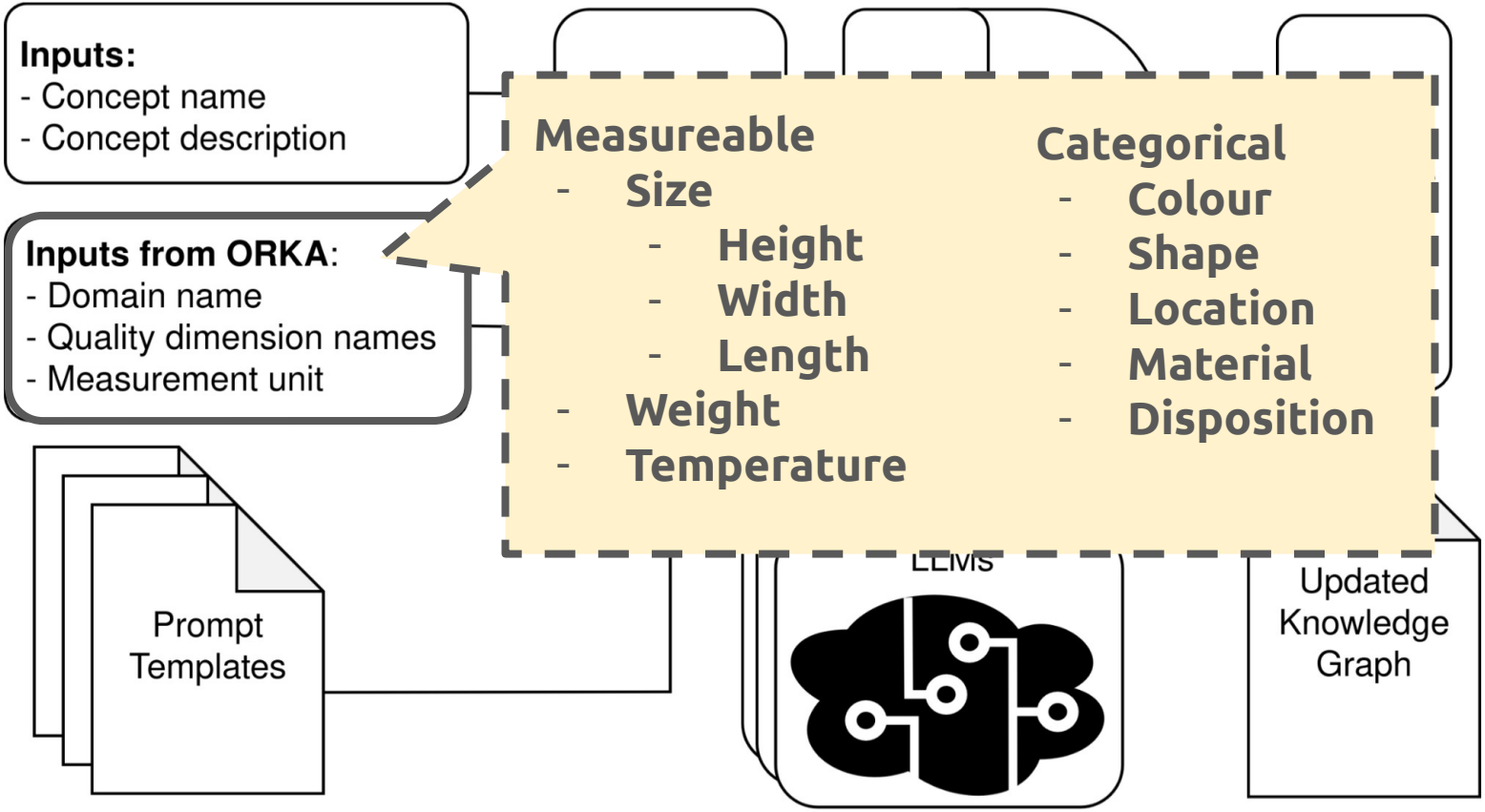
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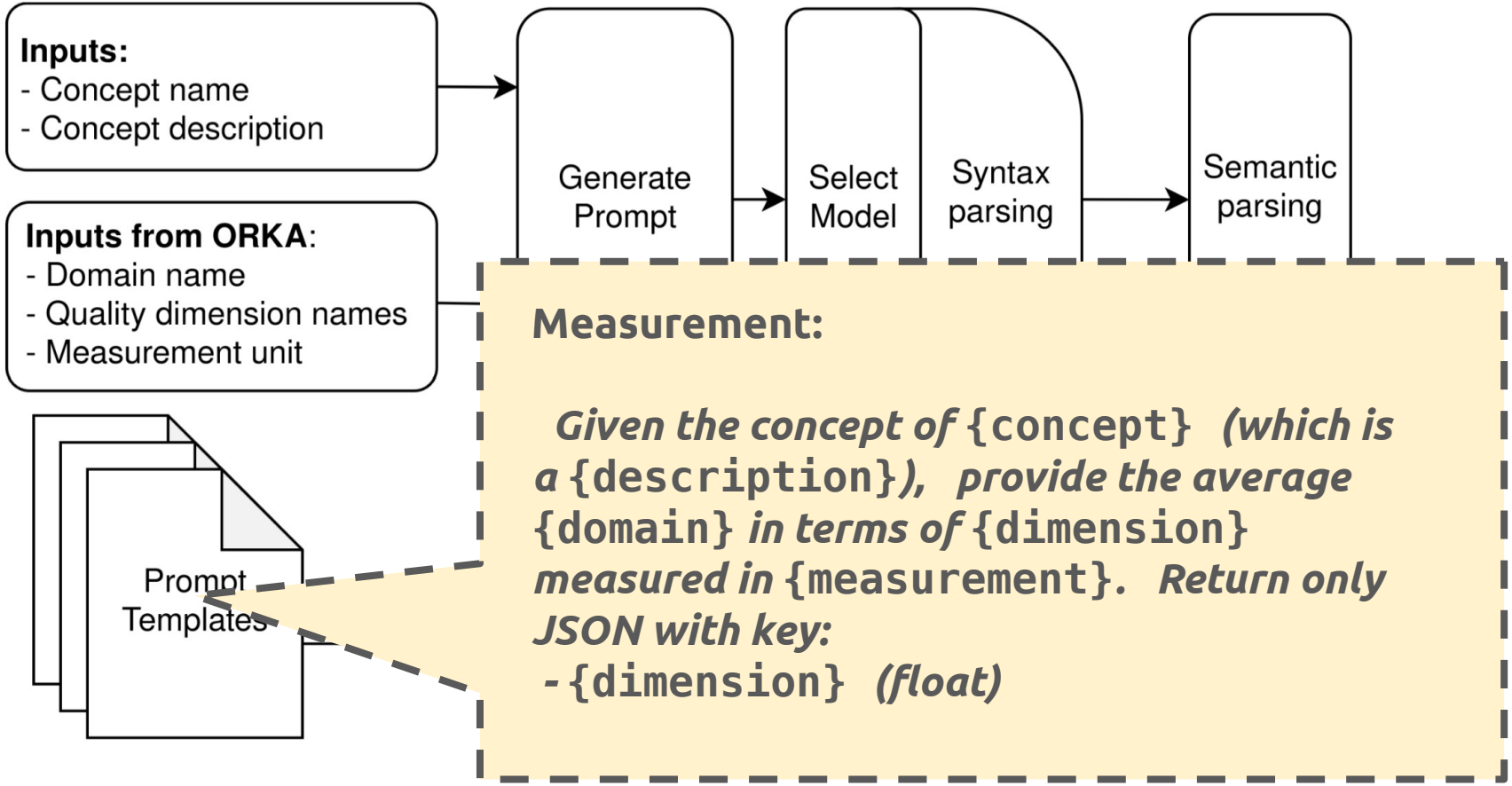
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Prompt Templates
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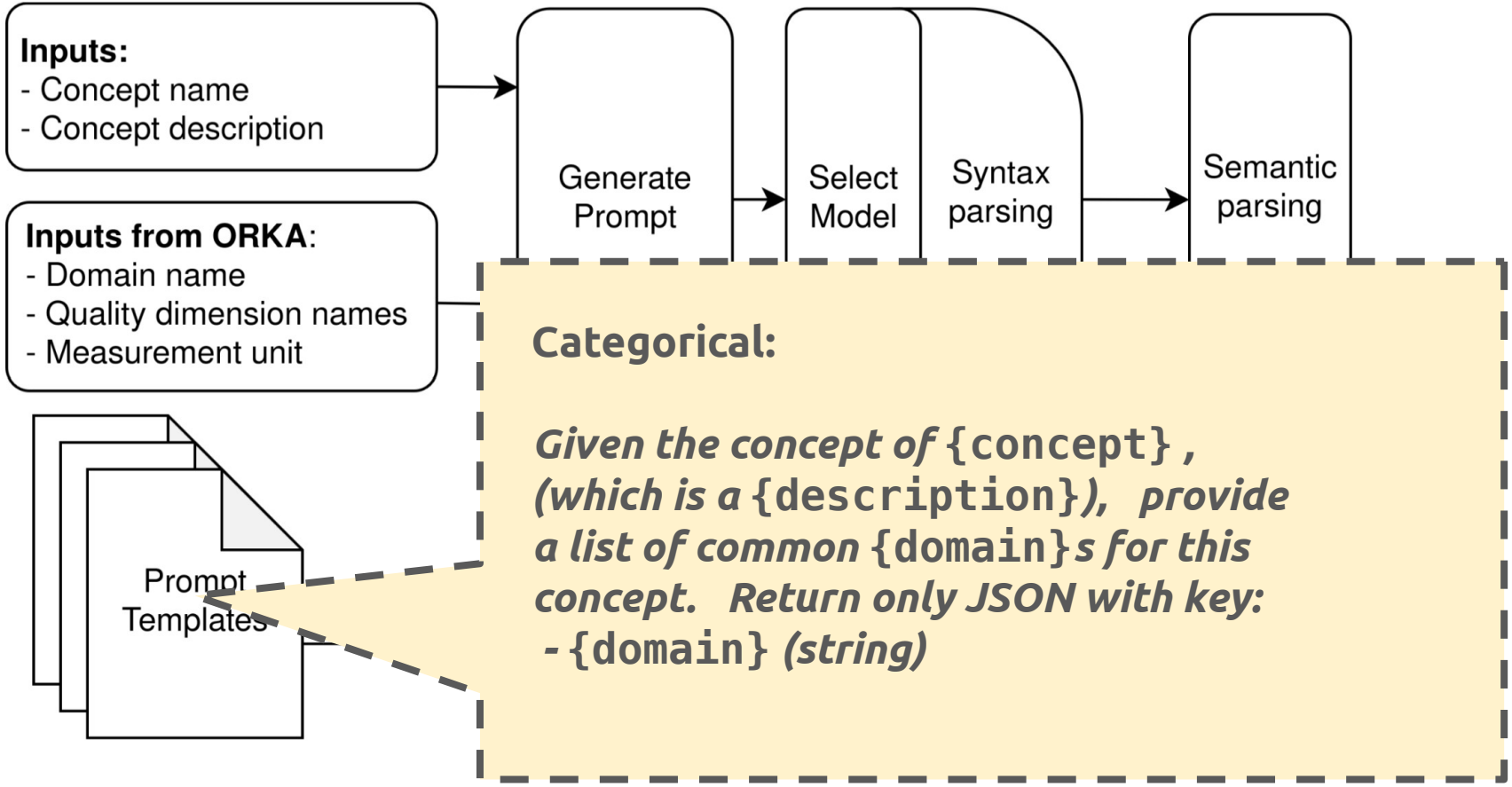
Syntax parsing
Semantic parsing
Measurement:
Given the concept of {concept} (which is a {description}), provide the average {domain} in terms of {dimension} measured in {measurement}. Return only JSON with key:
-{dimension} (float)

Pipeline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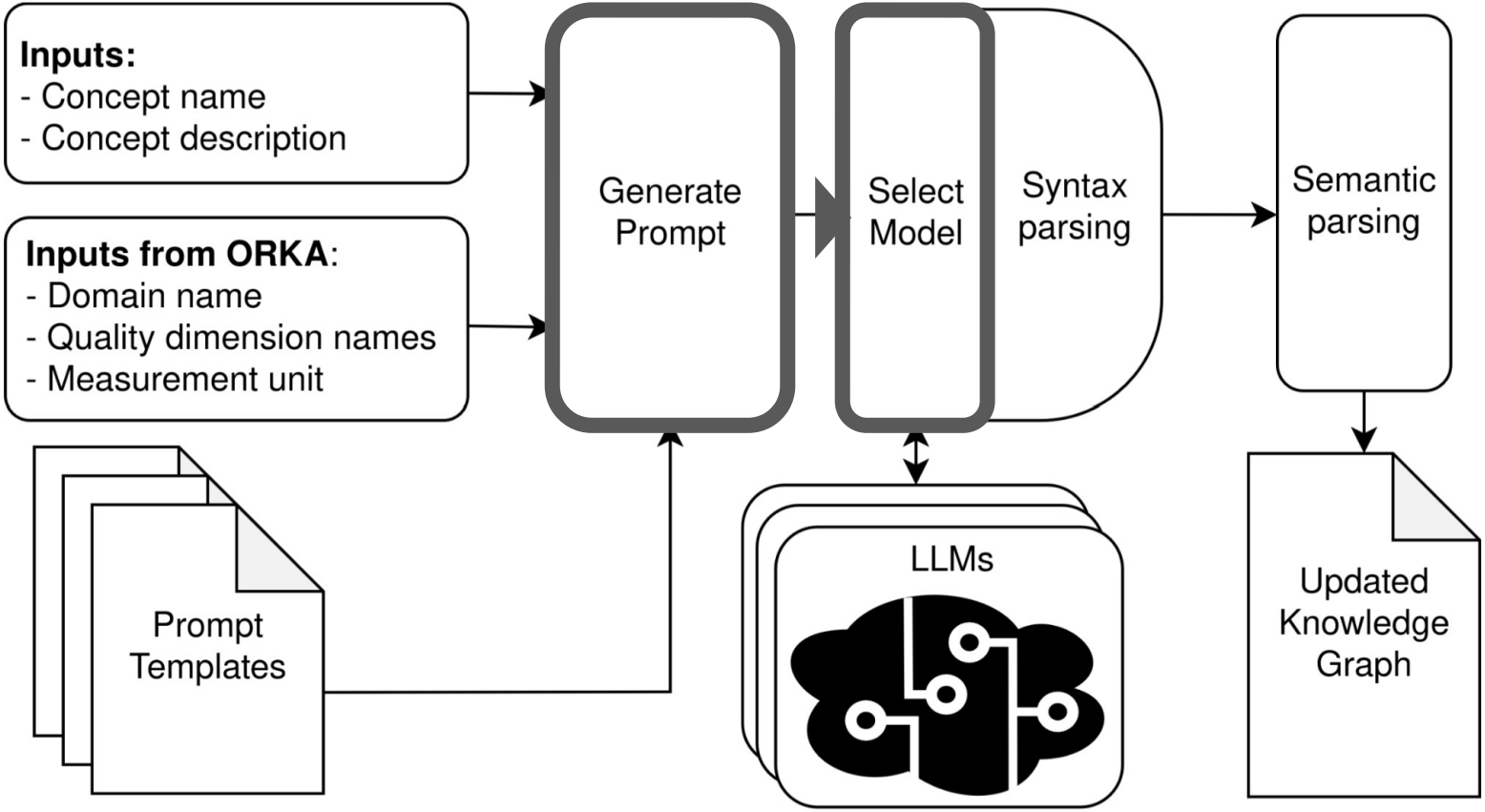
Categorical:
Given the concept of {concept} , (which is a {description}), provide a list of common {domain}s for this concept. Return only JSON with key:
- {domain} (string)

Pipeline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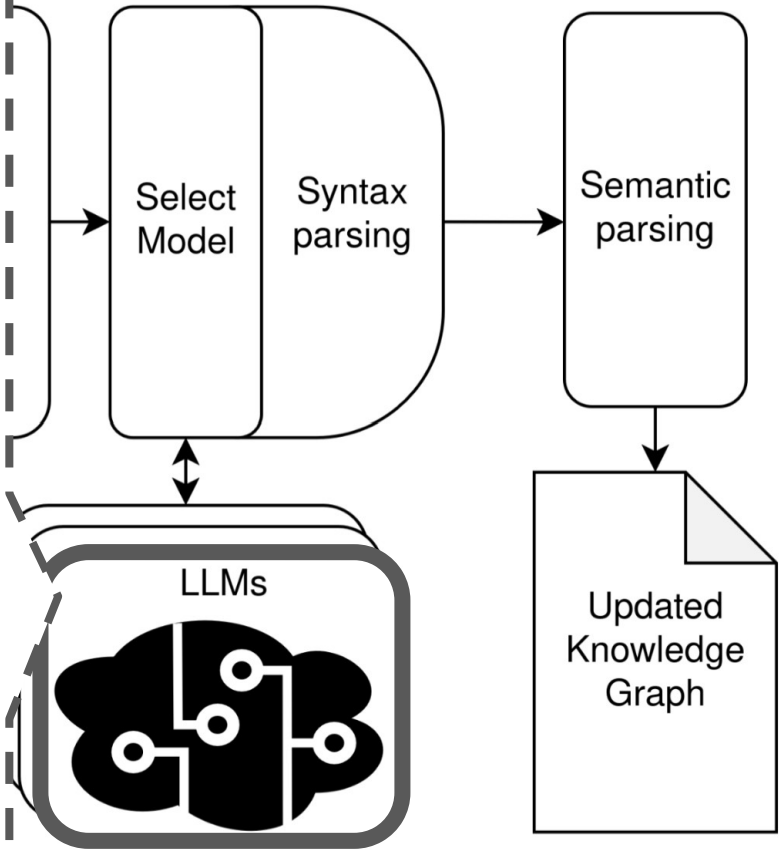
Pipeline

Model	Version	Params.	Developer
LLama	Standard	30b	Meta
LLava	Standard	7b	Meta
Phi3 mini (fp16)	Instruct	3.8b	Microsoft
Phi3 mini (q4)	Instruct	3.8b	Microsoft
Qwen2.5	Standard	1b	Alibaba Cloud
Qwen2.5	Standard	7b	Alibaba Cloud
Olmo2	Instruct	13b	AllenAI
LLama3.1	Instruct	8b	Meta
LLama3.1	Standard	8b	Meta
LLama3.1	Instant	8b	Meta
LLama4 Scout	Instruct	17b	Meta
Deepseek R1	Standard	1.5b	DeepSeek AI
Deepseek R1	Standard	8b	DeepSeek AI
Deepseek R1	Instruct	8b	DeepSeek AI
Deepseek R1	Distill	70b	DeepSeek AI
GPT-4.1	Standard	-	OpenAI



Pipeline

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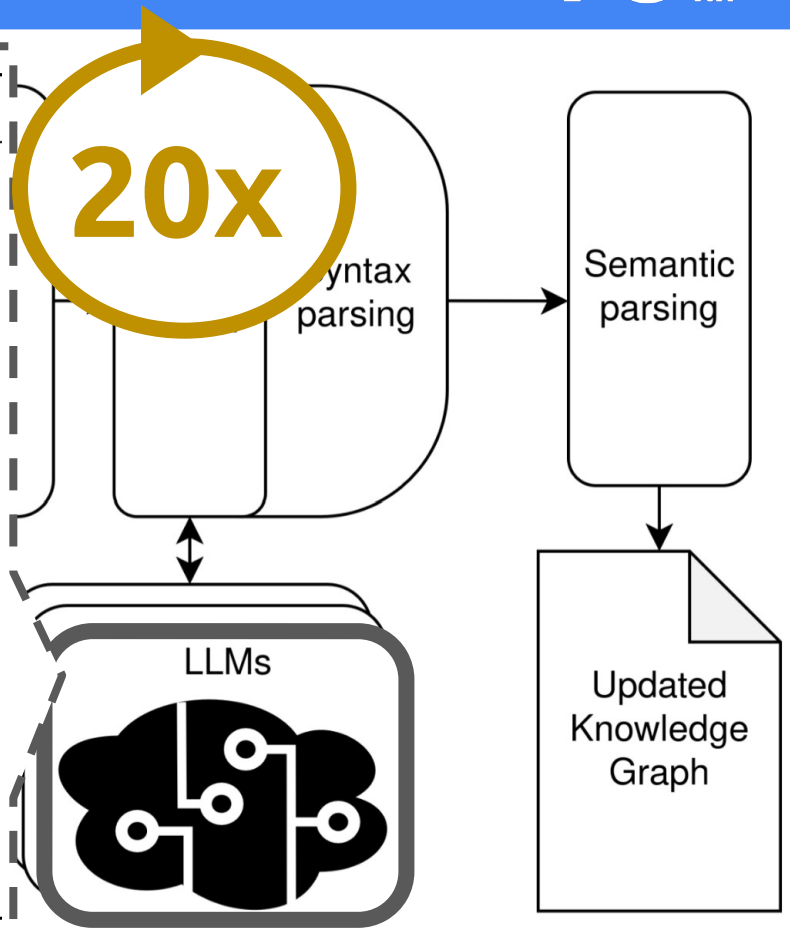
Model	Version	Params.	Developer
LLama	Standard	30b	Meta
LLava	Standard	7b	Meta
Phi3 mini (fp16)	Instruct	3.8b	Microsoft
Phi3 mini (q4)	Instruct	3.8b	Microsoft
Qwen2.5	Standard	1b	Alibaba Cloud
Qwen2.5	Standard	7b	Alibaba Cloud
Olmo2	Instruct	13b	AllenAI
LLama3.1	Instruct	8b	Meta
LLama3.1	Standard	8b	Meta
LLama3.1	Instant	8b	Meta
LLama4 Scout	Instruct	17b	Meta
Deepseek R1	Standard	1.5b	DeepSeek AI
Deepseek R1	Standard	8b	DeepSeek AI
Deepseek R1	Instruct	8b	DeepSeek AI
Deepseek R1	Distill	70b	DeepSeek AI
GPT-4.1	Standard	-	OpenAI



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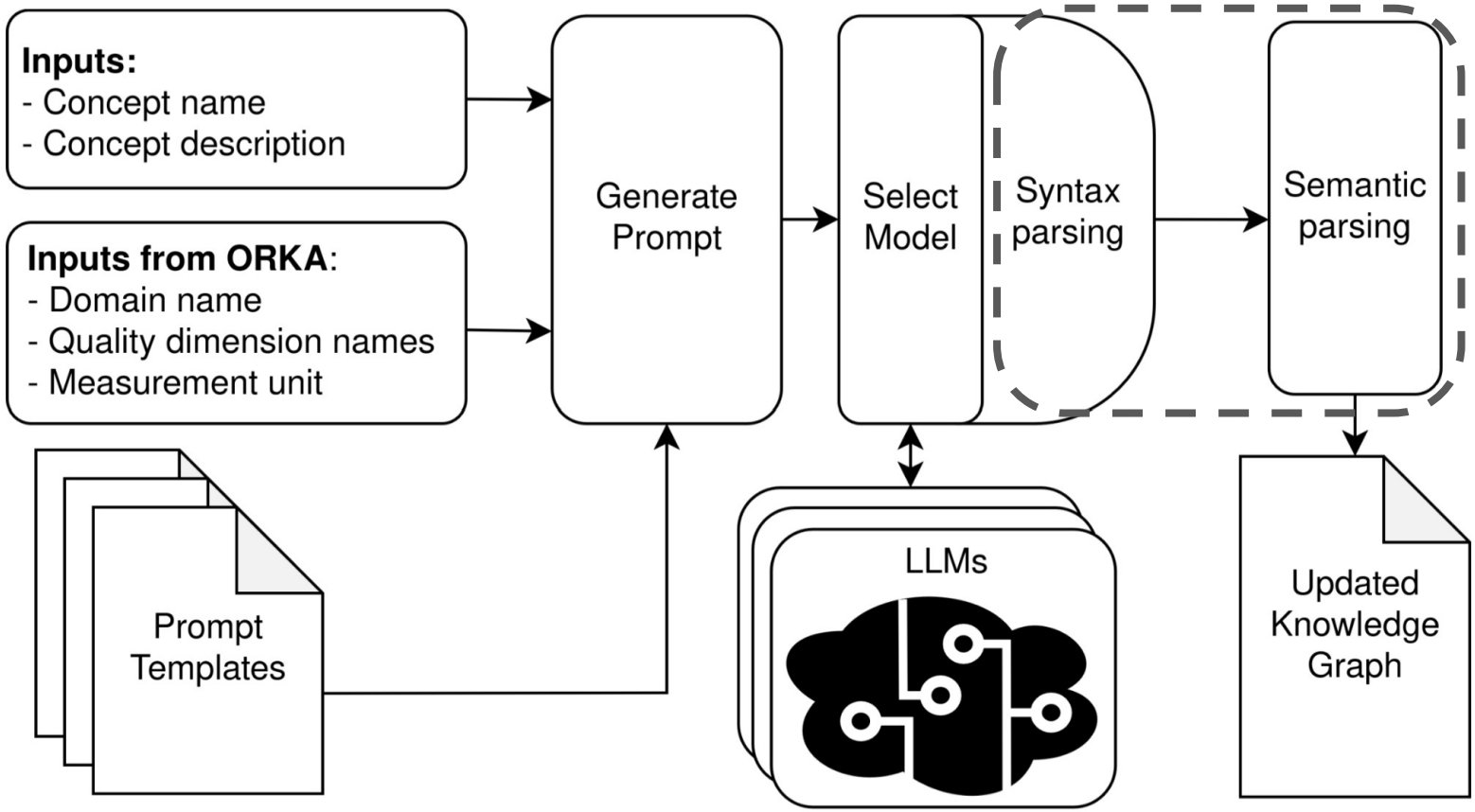
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Ground Truth

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Concepts: 80 objects of MS COCO

Annotation: 10 annotators for each object and each quality dimension (Amazon MTurk)

Post-processing: semi-manual

Instructions

Summary | **Detailed Instructions** | Examples

How to answer

- Use common sense; no research needed.
- If unsure, give your best guess.
- For **categorical** questions, type a comma-separated list (e.g. “red, black”), no need to type in more than five, but aim at three if possible. If none apply, write 'not applicable' (use na for short).
- For **measurable** questions:
  - **Mean** = the value you expect most of the time.
  - **Typical variation** = how far it usually strays above or below the mean (about ±1 SD).
  - Type a number or na if it doesn’t make sense for this object.
  - Select the units that fit best, or choose “n/a”.
  - When it comes to matters of size, imagine a bounding box around the object to provide the values (width, height, length).

Object: ${object_name} (${object_description})

Categorical properties

Typical colors of a ${object_name}:

e.g., red, blue, silver or na

Typical shapes of a ${object_name}:

e.g., rectangular, cylindrical or na

Typical locations where you find a ${object_name}:

e.g., kitchen, garage or na

Typical textures of a ${object_name}:

e.g., smooth, rough or na

Typical materials of a ${object_name}:

Measurable properties - Inputs only accept numbers or 'na'

Length of a ${object_name}

Mean (average):

number or na

Typical variation:

± value or na (± is implicit, not required to type)

○ m ○ cm ○ mm ○ in ○ ft ○ not applicable

Evaluation Method

$$s_m^{\text{num}} = \frac{1}{C} \sum_{c=1}^{C} \left(\frac{1}{D_{\text{num}}} \sum_{d=1}^{D_{\text{num}}} \frac{1}{1 + \left| \overline{x}_{c,d}^{m} - \overline{g}_{c,d} \right|} \right)$$

$$s_m^{\text{cat}} = \frac{1}{C} \sum_{c=1}^{C} \left(\frac{1}{D_{\text{cat}}} \sum_{d=1}^{D_{\text{cat}}} \frac{\left| P_{c,d}^{m} \cap G_{c,d} \right|}{\left| P_{c,d}^{m} \cup G_{c,d} \right|} \right)$$

Evaluation Method

Numerical values: score based on inverse error

$$s_m^{\text{num}} = \frac{1}{C} \sum_{c=1}^{C} \left(\frac{1}{D_{\text{num}}} \sum_{d=1}^{D_{\text{num}}} \frac{1}{1 + \left| \overline{x}_{c,d}^{m} - \overline{g}_{c,d} \right|} \right)$$

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Evaluation Method

D_{num}: numerical quality dimensions

C : concepts

$x^m_{c,d}$: model prediction

$g_{c,d}$: ground truth

$$s_m^{\text{num}} = \frac{1}{C} \sum_{c=1}^{C} \left(\frac{1}{D_{\text{num}}} \sum_{d=1}^{D_{\text{num}}} \frac{1}{1 + \left| \overline{x}_{c,d}^m - \overline{g}_{c,d} \right|} \right)$$

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Evaluation Method

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Categorical values: score based on jaccard similarity index

Evaluation Method

$$s_m^{\text{num}} = \frac{1}{C} \sum_{c=1}^{C} \left(\frac{1}{D_{\text{num}}} \sum_{d=1}^{D_{\text{num}}} \frac{1}{1 + \left| \overline{x}_{c,d}^{m} - \overline{g}_{c,d} \right|} \right)$$

$$s_m^{\text{cat}} = \frac{1}{C} \sum_{c=1}^{C} \left(\frac{1}{D_{\text{cat}}} \sum_{d=1}^{D_{\text{cat}}} \frac{\left| P_{c,d}^{m} \cap G_{c,d} \right|}{\left| P_{c,d}^{m} \cup G_{c,d} \right|} \right)$$

D_{cat}: categorical quality dimensions

$P^{m}_{c,d}$: model prediction (set of concepts)

Results

Results Overview

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Model	Version	Params.	Developer	E_{syn}	E_{sem}	s_{cat}	s_{exact}	s_{num}	E_{MRE}	σ_{MRE}
LLama	Standard	30b	Meta	0.73	13.99	0.1330	0.048	0.220	12.853	52.127
LLava	Standard	7b	Meta	1.42	7.02	0.2750	0.083	0.342	7.328	47.088
Phi3 mini (fp16)	Instruct	3.8b	Microsoft	**0.01**	4.80	0.3288	0.065	0.444	1.345	5.561
Phi3 mini (q4)	Instruct	3.8b	Microsoft	**0.01**	7.76	0.2468	0.043	0.451	**1.305**	6.130
Qwen2.5	Standard	1b	Alibaba Cloud	0.35	5.33	0.2702	0.053	0.309	7.003	40.323
Qwen2.5	Standard	7b	Alibaba Cloud	0.02	5.72	0.3504	0.078	0.512	1.500	8.037
Olmo2	Instruct	13b	AllenAI	0.02	13.83	0.3524	0.085	0.417	5.011	39.197
LLama3.1	Instruct	8b	Meta	0.08	6.18	0.3159	0.058	0.419	2.219	10.505
LLama3.1	Standard	8b	Meta	1.69	5.56	0.3608	**0.090**	0.411	2.369	10.368
LLama3.1	Instant	8b	Meta	1.21	6.57	0.3283	0.055	0.388	4.610	24.555
LLama4 Scout	Instruct	17b	Meta	**0.01**	3.54	**0.3922**	0.088	0.580	1.517	8.616
Deepseek R1	Standard	1.5b	DeepSeek AI	10.56	17.54	0.0697	0.018	0.206	11.730	50.824
Deepseek R1	Standard	8b	DeepSeek AI	0.19	8.72	0.2680	0.055	0.378	2.748	13.469
Deepseek R1	Instruct	8b	DeepSeek AI	0.51	7.26	0.2082	0.043	0.415	2.679	19.315
Deepseek R1	Distill	70b	DeepSeek AI	34.25	3.67	0.3295	0.088	0.502	1.799	11.505
GPT-4.1	Standard	-	OpenAI	**0.01**	**2.27**	0.3796	0.065	**0.594**	2.125	13.113

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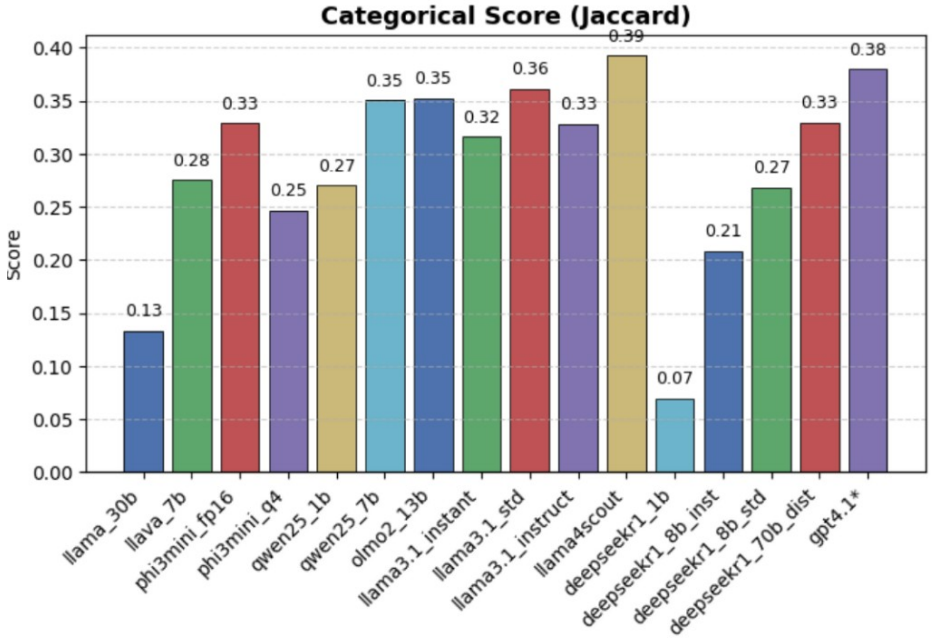
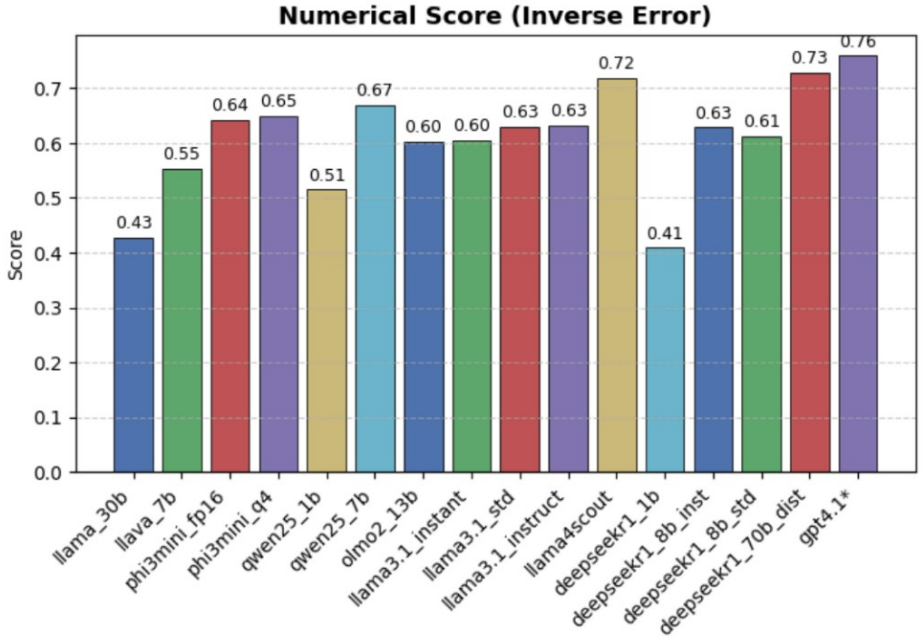
Results - Visualized





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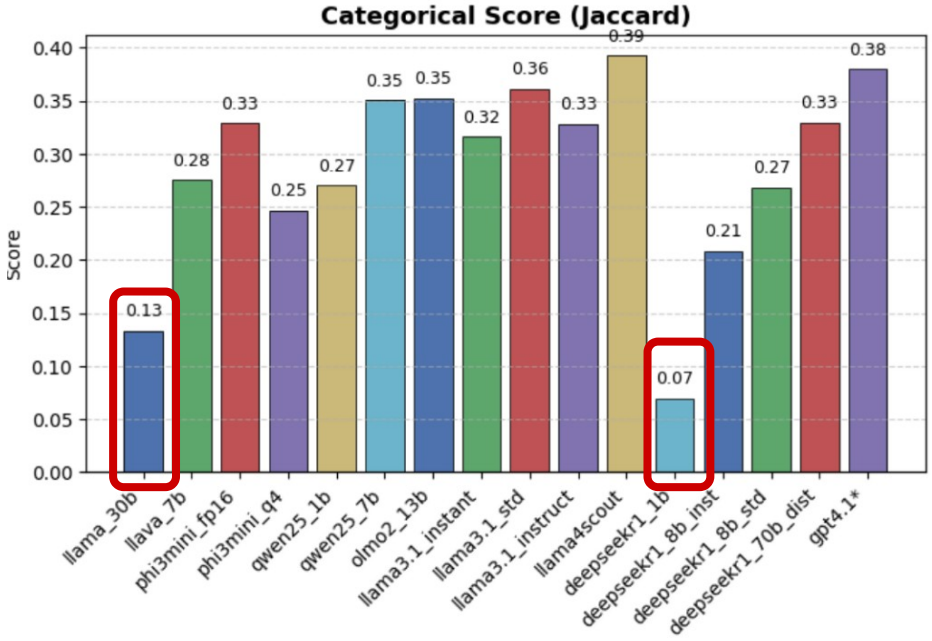

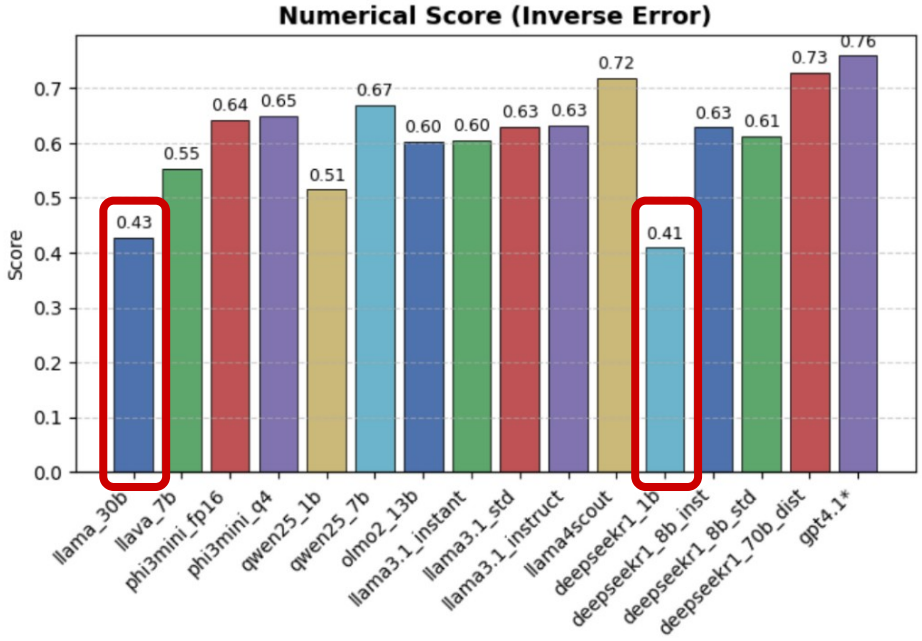

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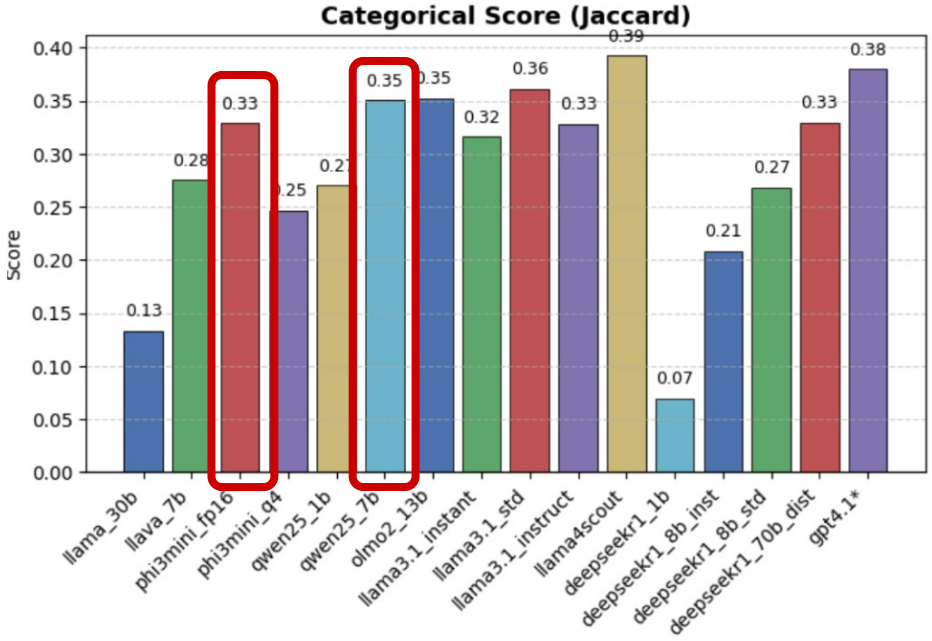
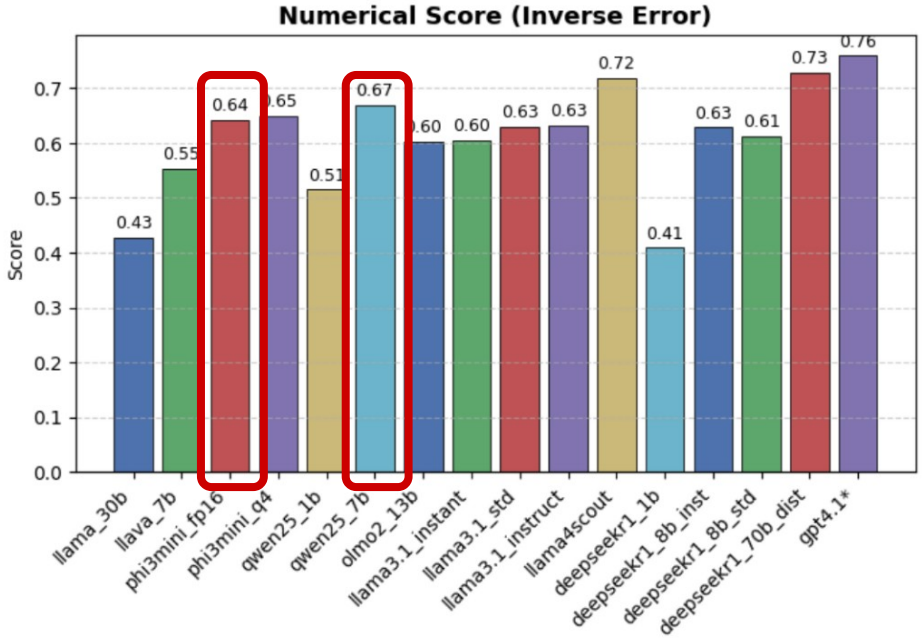
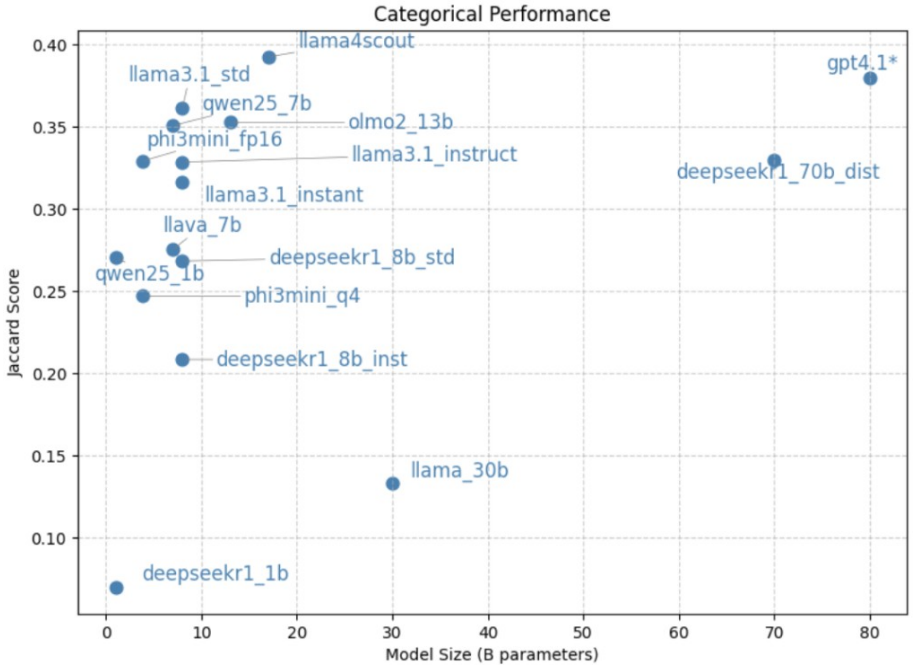
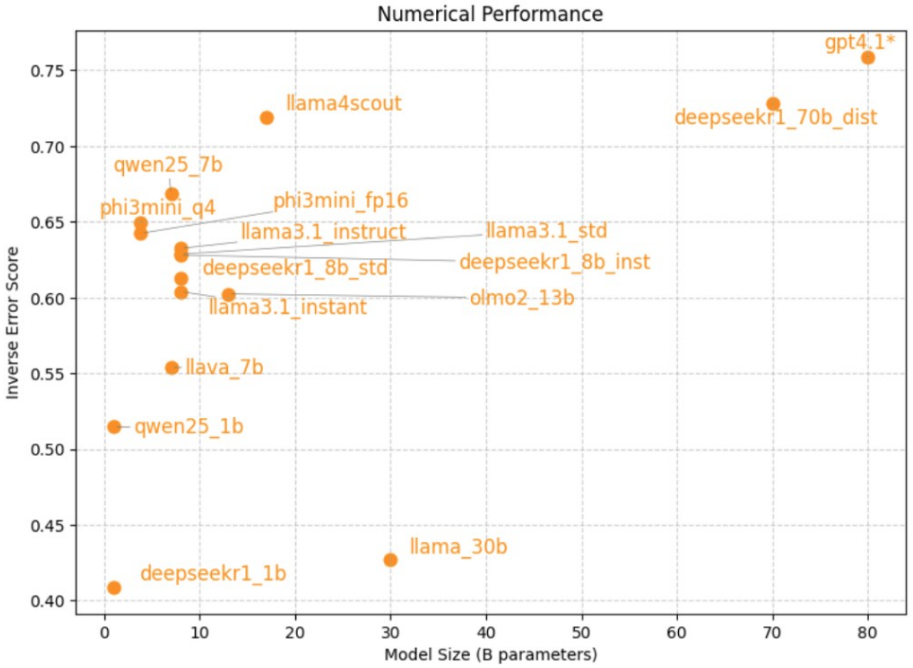




Results - Grouped by model size





Takeaways and Challenges

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Takeaways:

- Size of model doesn't necessarily matter (as long as it's at least around 2b)
- More recent models are better
- Reasoning models not advised (disabled reasoning or large context window)
- Model version (instruct / standard / instant) doesn't make a big difference
- Measurement units: metric system more consistent, but not a bottleneck

Challenges:

- Representing some dimensions is challenging (e.g. location, shape)
- How do we determine the context vector?
- How to represent distributions and ranges of values in KGs?
- How do we evaluate the quality of the resulting knowledge? (semantic parsing)
- How to get reliable ground truth? Are LLMs better then MTurk?

Next Steps

Future (current) work

- Extend pipeline to ImageNet1000 (Imagenet-CS)
- Examine linking performance to Imagenet-CS
- Downstream task: Improve task planning*
- Examine the quality of MTurk vs. LLM annotations
- Use pipeline for dimensional models of emotions

*Adamik, M., & Forte, P. (2025, November). PlanOwl: Automated PDDL Files Generation from OWL Ontologies and Visual Language Models. In *Proceedings of the AAAI Symposium Series*

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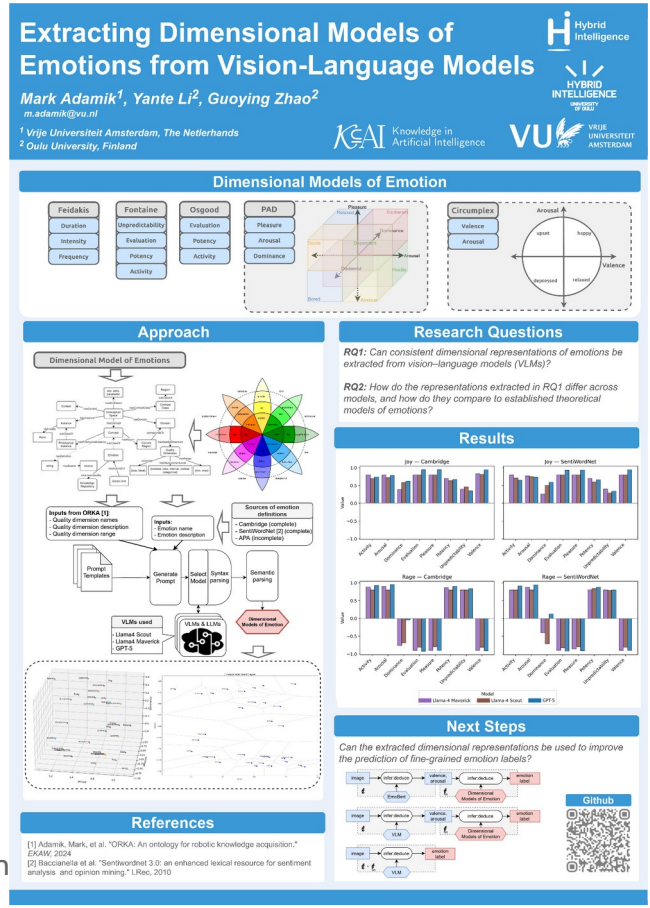

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Thank You!

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KCAI Knowledge in Artificial Intelligence